

J = USVT

$\tilde{j} = \tilde{U} \tilde{S} \tilde{V}^T$

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11

$x_1^{l-1} f$

1

1

11

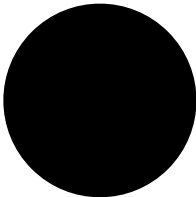
$x_1^l f$

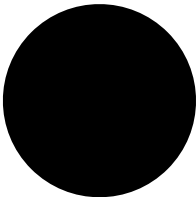
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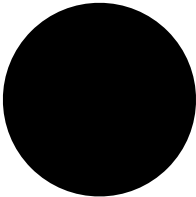
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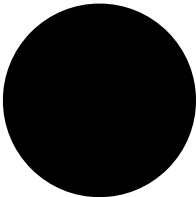
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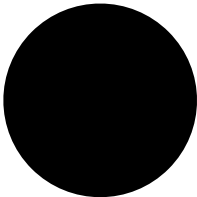
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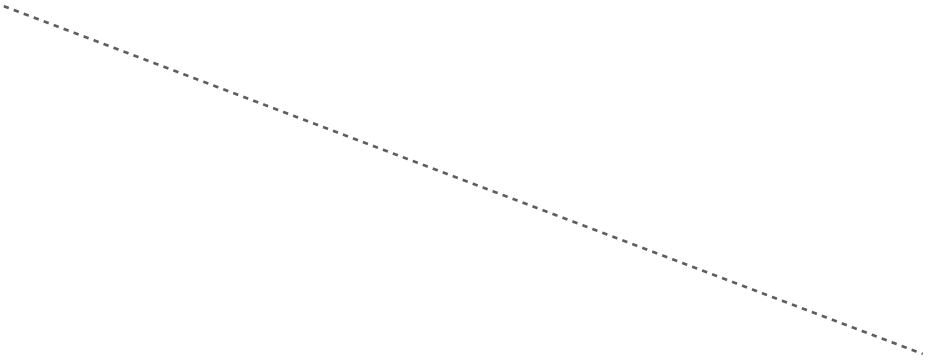


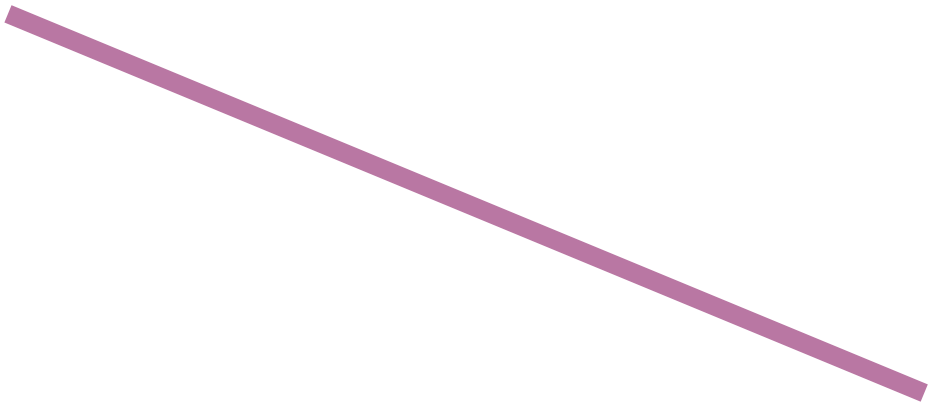


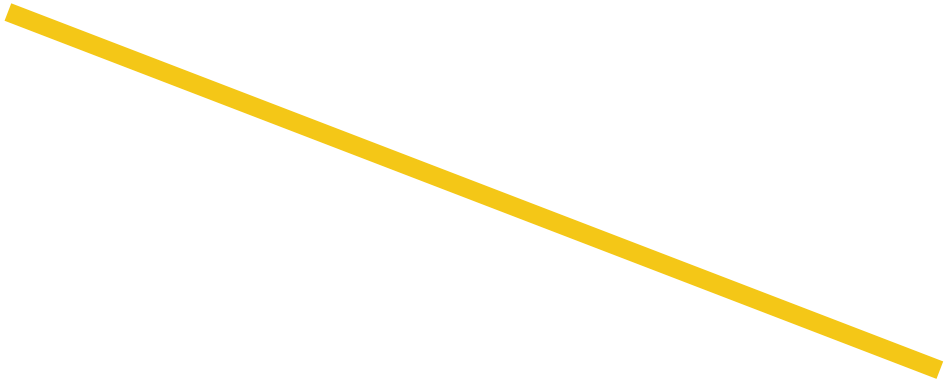


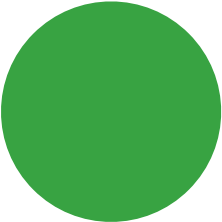


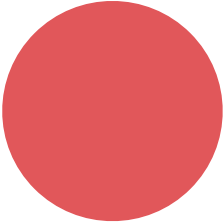
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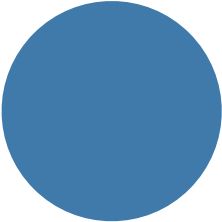


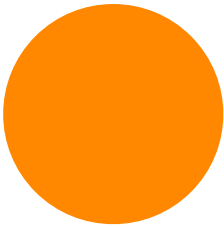












11

x_1^{l-1}

f

1

22

x_2^l

f

2

12

12

x_1^{l-1} f

2

34

x_3^l f

4

1234

12

x_1^{l-1} f

2

34

x_3^l f

4

1234

Constructing J

$$J(X) = \frac{\partial f}{\partial X}(X)$$

Constructing A

$$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

$$A = \tilde{U}^T J \tilde{V}$$

Depth-wise Coupling

$$\left. \begin{aligned} J_{11}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_1 \\ J_{11}^{l+1} &= \frac{\partial}{\partial x_1^l} (f^{l+1}(X^l))_1 \end{aligned} \right\} A_{11}^{l,l+1}$$

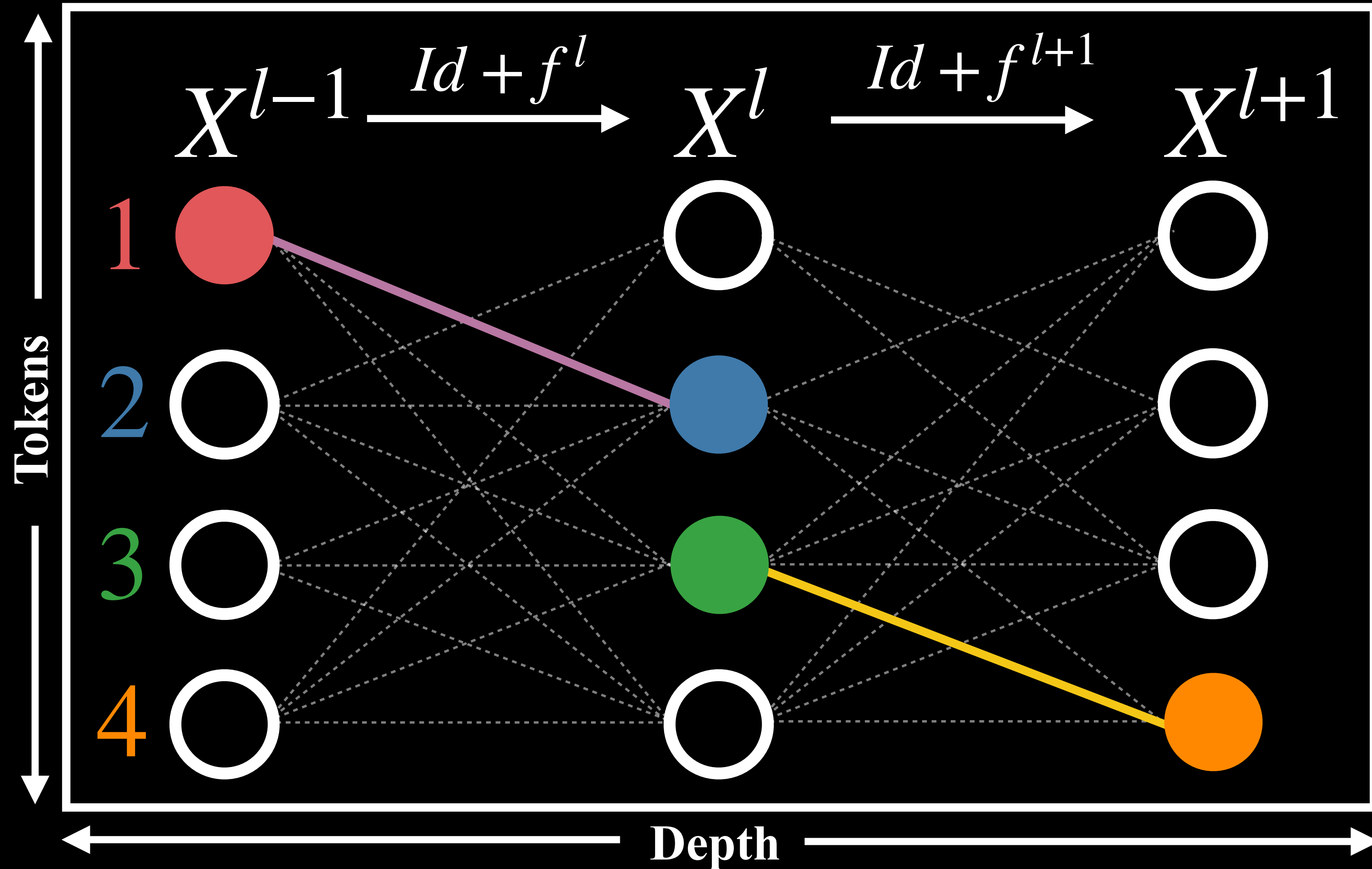
Token-wise Coupling

Self coupling

$$\left. \begin{aligned} J_{11}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_1 \\ J_{22}^{l+1} &= \frac{\partial}{\partial x_2^l} (f^{l+1}(X^l))_2 \end{aligned} \right\} A_{12}^{l,l+1}$$

Context coupling

$$\left. \begin{aligned} J_{12}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_2 \\ J_{34}^{l+1} &= \frac{\partial}{\partial x_3^l} (f^{l+1}(X^l))_4 \end{aligned} \right\} A_{1234}^{l,l+1}$$



Constructing J

$$J(X) = \frac{\partial f}{\partial X}(X)$$

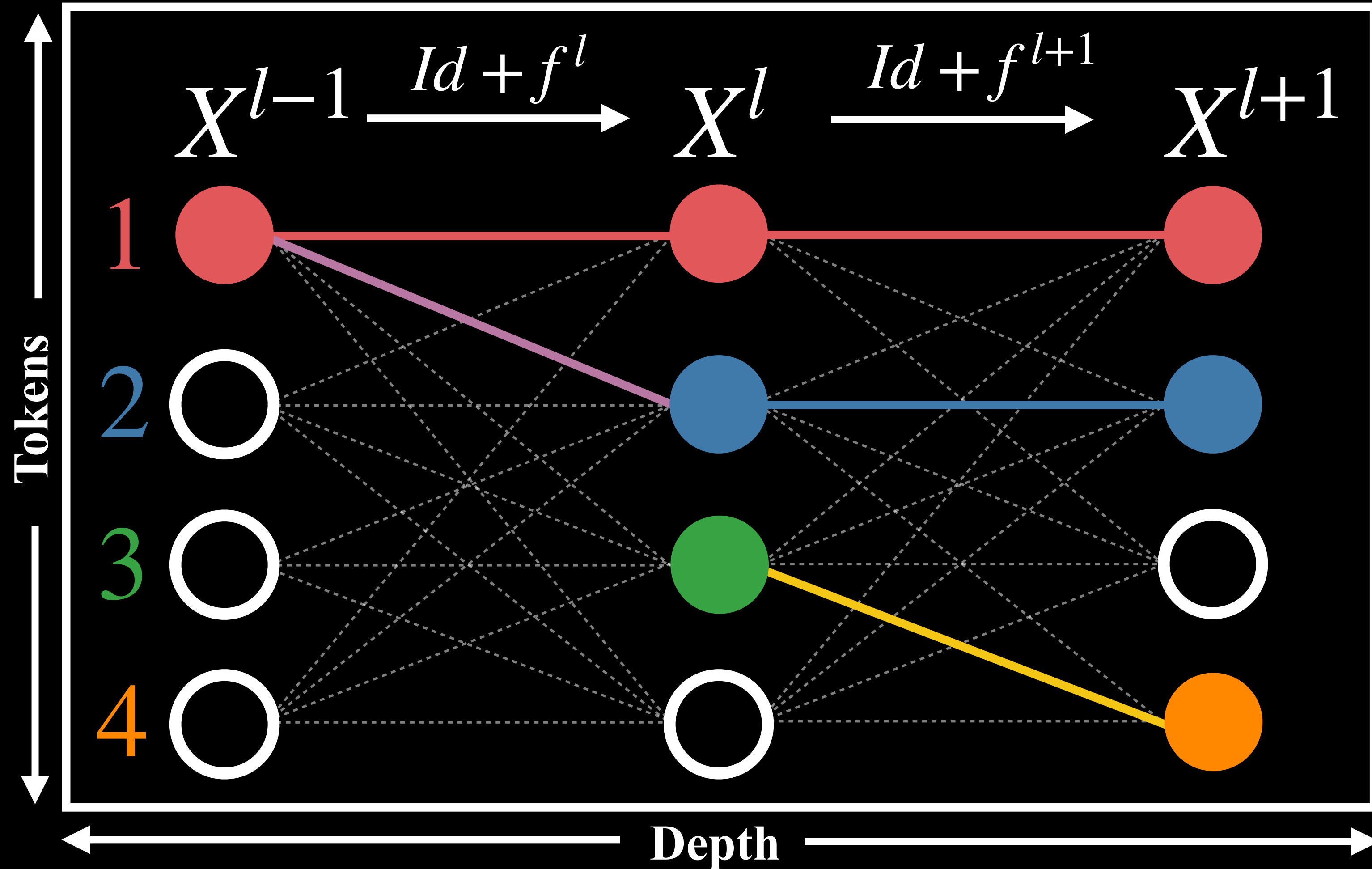
Constructing A

$$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

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Depth-wise Coupling

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Token-wise Coupling

Self coupling

$$\left. \begin{aligned} J_{11}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_1 \\ J_{22}^{l+1} &= \frac{\partial}{\partial x_2^l} (f^{l+1}(X^l))_2 \end{aligned} \right\} A_{12}^{l,l+1}$$

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